

Moles

* Relative atomic mass: The average mass of all naturally occurring atoms of an element compared to $\frac{1}{12}$ of C-12

Relative atomic mass = Nucleon mass

$$A_r = \frac{\sum \text{isotopic mass} \times \text{percentage abundance}}{100}$$

$$A_r = \frac{\sum \text{isotopic mass} \times \text{relative abundance}}{\text{Total relative abundance}}$$

* Relative molecular / formula mass: Sum of relative atomic mass of each element written within in a chemical formula

→ Relative molecular mass: Used for molecular compounds

→ Relative formula mass: Used for ionic compounds.

M_r : $X Y$

: $A_r \text{ of } X \times \text{atoms} + A_r \text{ of } Y \times \text{atoms}$

* Percentage composition by mass: It tells how much of each element the compound contains as a percentage of the total mass.

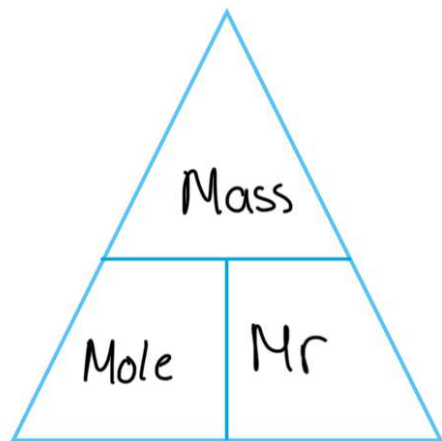
$$\% \text{ by mass} = \frac{A_r \text{ of the element} \times \text{Number of atoms}}{M_r \text{ of the compound}} \times 100\%$$

Avogadro's constant: The number of particles found in one mole: 6.02×10^{23}

Mole: The amount of a substance that contains 6.02×10^{23} atoms, ions or molecules

Molar mass: The mass of one mole of any substance

Molar mass = Relative atomic mass = Molecular mass
(Mr) (in grams) (in grams)



$$\rightarrow \text{Mole} = \frac{\text{Mass}}{\text{Mr}}$$

$$\rightarrow \text{Mass} = \text{Mole} \times \text{Mr}$$

$$\rightarrow \text{Mr} = \frac{\text{Mass}}{\text{Mole}}$$

Molecular Formula: The formula of a compound that specifies the actual number of atoms of each element in one molecule.

Empirical Formula: The formula of a compound with the smallest whole-number mole ratio of the elements.

Steps to write the empirical formula:

1- Find the mass of the elements that combine.

2- Calculate the number of moles: $\frac{\text{mass}}{M_r}$

3- Divide the number of moles by the smallest mole ratio.

4- If the mole ratio is not a whole number, multiply by a coefficient to get a whole number

5- Write the empirical formula with the simplified mole ratio.

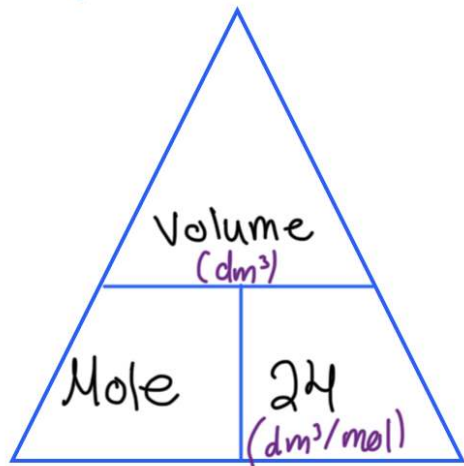
→ To find the molecular formula, find n , then multiply n by the empirical formula.

$$n = \frac{M_r \text{ of the compound (given in question)}}{\text{Mass of one mole of the empirical formula. (} M_r \text{ of empirical formula).}$$

To find a formula experimentally: example hydrated copper(II) sulfate.

* Add a known mass of hydrated CuSO_4 into a crucible, and weigh it. Heat the crucible for a specific amount of time, and allow it to cool, then re-weigh it again. Repeat the procedure until you reach constant mass.

* One mole of any gas at room temperature and pressure (r.t.p) always occupies 24 dm^3 , which is the molar volume of a gas.



$$\text{Mole} = \frac{\text{Volume}}{24}$$

$$\text{Volume} = \text{Mole} \times 24$$

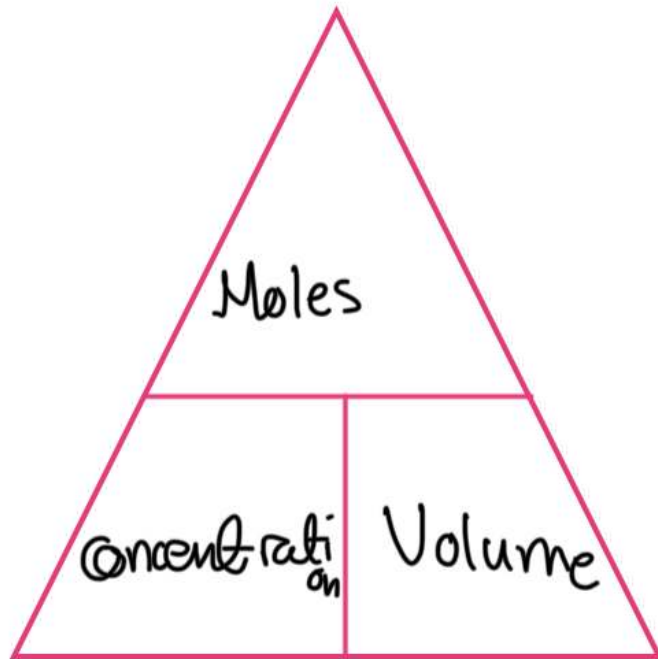
In reactions that involve only gases:

$$\frac{\text{Ratio of volume}}{\text{Ratio of moles}} =$$

Concentration of solution: The amount of solute dissolved in 1 dm^3 of solution.

Mass concentration of a solution is measured in grams per cubic decimeter (g/dm^3)

Molar concentration of a solution is measured in mole per cubic decimeter (mol/dm^3) = M



$$\rightarrow \text{Mole} = \text{Concentration} \times \text{Volume}$$

$$\rightarrow \text{Concentration} = \frac{\text{Mole}}{\text{Volume}}$$

$$\rightarrow \text{Volume} = \frac{\text{Mole}}{\text{Concentration}}$$

	Acid	Base	Neutral
Methyl orange	Red	Yellow	Orange
Thymolphthalein	Colorless	Blue	Colorless
Phenolphthalein	Colorless	Pink	Colorless

Steps to carry out acid-Base titration to know the concentration of an unknown acid solution:

1- Rinse the equipment with water to remove the residue from previous experiments. Then rinse the burette with acid, and the pipette with alkali to remove the remain of water. Rinse the conical flask with water only.

2- Add using a pipette 25cm^3 of alkali into the conical flask.

3- Add a few drops of an indicator.

4- Add using a burette the acid to the alkali, and make sure the jet space in the burette is filled with acid.

5- Swirl the mixture while adding the acid, and add the acid dropwise at end point.

6- Using a white tile, observe the color change.

7- Note the burette reading before and after the addition of acid.

8- Repeat the titration until at least two concordant results are obtained.

Errors

- 1- Overshoots, or missing end point.
- 2- Water left in burette / pipette
- 3- Air log below tap in burette
- Air log in pipette.
- 4- Funnel left on top of the burette.
- 5- Washing the flask with the solution that will go in it.

Improvements

- * Add the acid from the burette drop by drop near the end point.
- * Rinse the burette / pipette with the solution to be added in
- * Run some of the solution from the burette by titration.
- * Empty it and refill it again

Limiting reactant: The reactant that is not in excess, and it controls when the reaction ends.

Excess reactant: The reactant that is left over unreacted at the end of the reaction.

How to know the limiting reactant?

1- Calculate, using the given information, the moles of each reactant.

2- Mole ratio: First reactant: Product

Second reactant: Product

3- The reactant that produced the smallest mole of product is the limiting reactant.

How to calculate the mass of product?

* Using the limiting reactant, find the moles of the product, and multiply by the molar mass

How to calculate the mass of unreacted reactant?

1- Use the limiting reactant & the mole ratio to work out the moles used from excess reactant.

2- Work out the mass by multiplying the moles by molar mass

3- Find the mass left by subtracting the mass of the reactant from the total mass.

Why would there be yield loss during a reaction?

1- The reaction may not be totally complete.

2- Errors may be made in weighing the reactants or the products.

3- Materials may be lost in carrying out the reaction, or in transferring and separating the products.

4- During crystallization some remains in the solution.

Actual yield: The amount of product found in a non-complete reaction. (Given in the question)

Theoretical yield: The maximum yield predicted by equation.

→ Actual yield is less than theoretical yield.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

$$\text{Percentage purity} = \frac{\text{Mass of pure substance (calculated)}}{\text{Mass of impure substance (given)}} \times 100\%$$

$$\rightarrow \text{Number of particles} = \text{Moles} \times \text{Avogadro's constant } 6.02 \times 10^{23}$$

$$\rightarrow \text{Moles} = \frac{\text{Number of particles}}{6.02 \times 10^{23}}$$